OmicsLite BRSP | Assignment Week 4

Differential Expression Genes Analysis in Salt-Stressed and Normal *Arabidopsis thaliana* Transcriptomics Data

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A. Introduction

One of the pressing issues in food production is salt intrusion due to the climate crisis. The seawater level that keeps increasing because of global warming can contaminate the agricultural land located close to the shore, especially in an archipelago state. While controlling salt intrusion in freshwater is complex, investigating the underlying mechanism of plant transcriptional response to saline conditions can open up an opportunity to develop plant salt-tolerance traits. Many researchers have already investigated the transcriptional response of plants in saline conditions. One of them is Dinneny et al. (2008), with *Arabidopsis thaliana* ecotype Columbia (Col-0) as their model organism. *A. thaliana* is widely used in basic plant research because it has a simple genome that was already fully sequenced at the end of 2000 (Meinke et al., 1998). Its life cycle can be completed in 6 weeks, and the small full growth size makes it versatile and convenient to be grown in a petri dish or pot. The Columbia ecotype is one of the accepted standard model organisms for genetic and molecular studies.

NCBI Gene Expression Omnibus (GEO) is one of the public data repositories whose database can be mined to conduct differential gene expression (DEG) analysis from various transcriptomics research. The microarray data from Dinneny et al. (2008) can be found with the accession number of GSE7642. There is a data analysis tool available embedded in the NCBI GEO web with the name GEO2R. However, its capacity is only limited to analyze microarray data and provide basic plots. Therefore, the R statistical software was used as a robust alternative. R can analyze other types of transcriptomics data and generate various data visualizations. R is an open-source language and software environment developed for statistical computing, data analysis, and graphics (R Core Team, 2025).

In this study, the GSE7642 dataset will be analyzed for differential expression between salt-stressed and normal conditions with R, and the gene ontology of the significant genes will be determined.

B. Methods

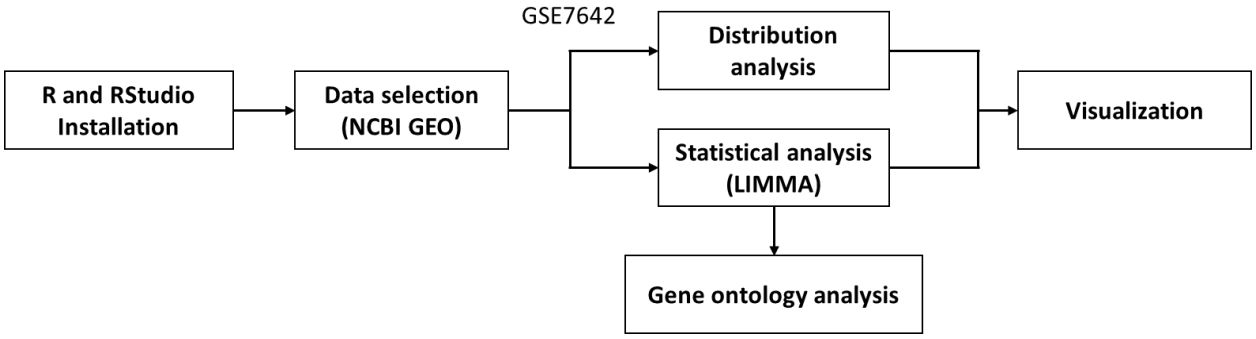

Figure 1. The workflow of this study.

Differential expression analysis was performed using the R statistical software version 4.5.2 (R Core Team, 2025) within RStudio version 20206.01.1+403 (Posit Team, 2025). BiocManager (Morgan & Ramos, 2025) was installed to manage the Bioconductor package.

The microarray dataset with the accession ID GSE7642 was retrieved from the GEO database using the GEOquery package (Davis & Meltzer, 2007), which was later preprocessed by log2 transformation. The GPL198 platform [ATH1121501] Affimetrix Arabidopsis ATH1 Genome Array was used in the GSE7642 dataset. The GSE7642 dataset contains 12 sample microarrays from *A. thaliana* whole roots, including 2 replication in standard condition and 2 replication for each 5 time points with 140 mM NaCl treatment. Six sample groupings (none, X30.minutes, X1.hour, X4.hours, X16.hours, and X32.hours) were analyzed for DEGs using the LIMMA (Linear Model for Microarray Data) package (Ritchie et al., 2015). The data was filtered using a fold change of |Log2FC| > 1 and an adjusted p-value of 0.05 calculated with the Benjamini & Hochberg (Benjamini & Hochberg, 1995) false discovery rate (FDR). Gene annotations were mapped according to the ath1121501.db database (Carlson, 2021) using AnnotationDbi (Pagès et al., 2024). Data manipulation and visualization were carried out using dplyr (Wickham et al., 2023), ggplot2 (Wickham, 2016), and pheatmap (Kolde, 2019). The top 50 DEGs were submitted to ShinyGo ver. 0.85 (Ge et al., 2020) to determine their gene ontology with *A. thaliana* araport11 database.

C. Result and Discussion

From the GSE7642 dataset, we compared 5 different time points of saline treatment to the control (Figure 2). The result showed that the peak of DEGs in comparison to normal conditions happened when A. thaliana was exposed to saline conditions for 4 hours with a total of 824 DEGs. After 4 hours, the number of genes that expressed differently slowly decreased. At all time points, the number of upregulated genes was higher than downregulated genes under salt stress conditions. The detailed number of DEGs for each comparison was presented in Table 1.

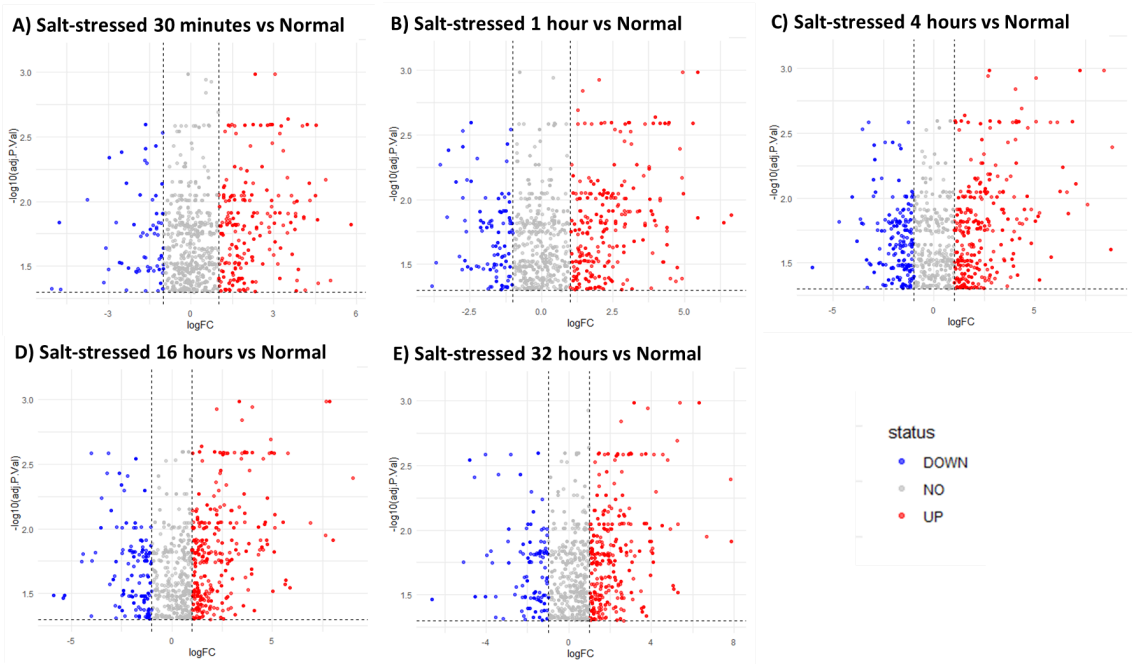

Figure 2. Volcano plot of DEGs in salt-stressed *A. thaliana* in five time points compared to control from the GSE7642 dataset. A to E is the visual representation of the amount of DEGs that are downregulated (in blue) and upregulated (in red) over time when left in saline for 30 minutes (A), 1 hour (B), 4 hours (C), 16 hours (D), and 32 hours (E).

Table 1. The number of DEGs from five time points of salt-stressed *A. thaliana* compared to control from the GSE7642 dataset. The number of upregulated and downregulated DEGs was shown from the total DEGs.

Comparison	DEGs	Upregulated	Downregulated
30 minutes vs Normal	463	371	92
1 hour vs Normal	600	446	156
4 hours vs Normal	824	530	294
16 hours vs Normal	750	530	220
32 hours vs Normal	664	488	176

The top 50 DEGs were shown in a heatmap with the annotated genes (Figure 3). There are four GPL96 probe IDs without annotated genes detected in the ath1121501.db database (256114_at, 267080_at, 253344_at, and 250624_at). Therefore, the four probe IDs were removed in gene ontology analysis.

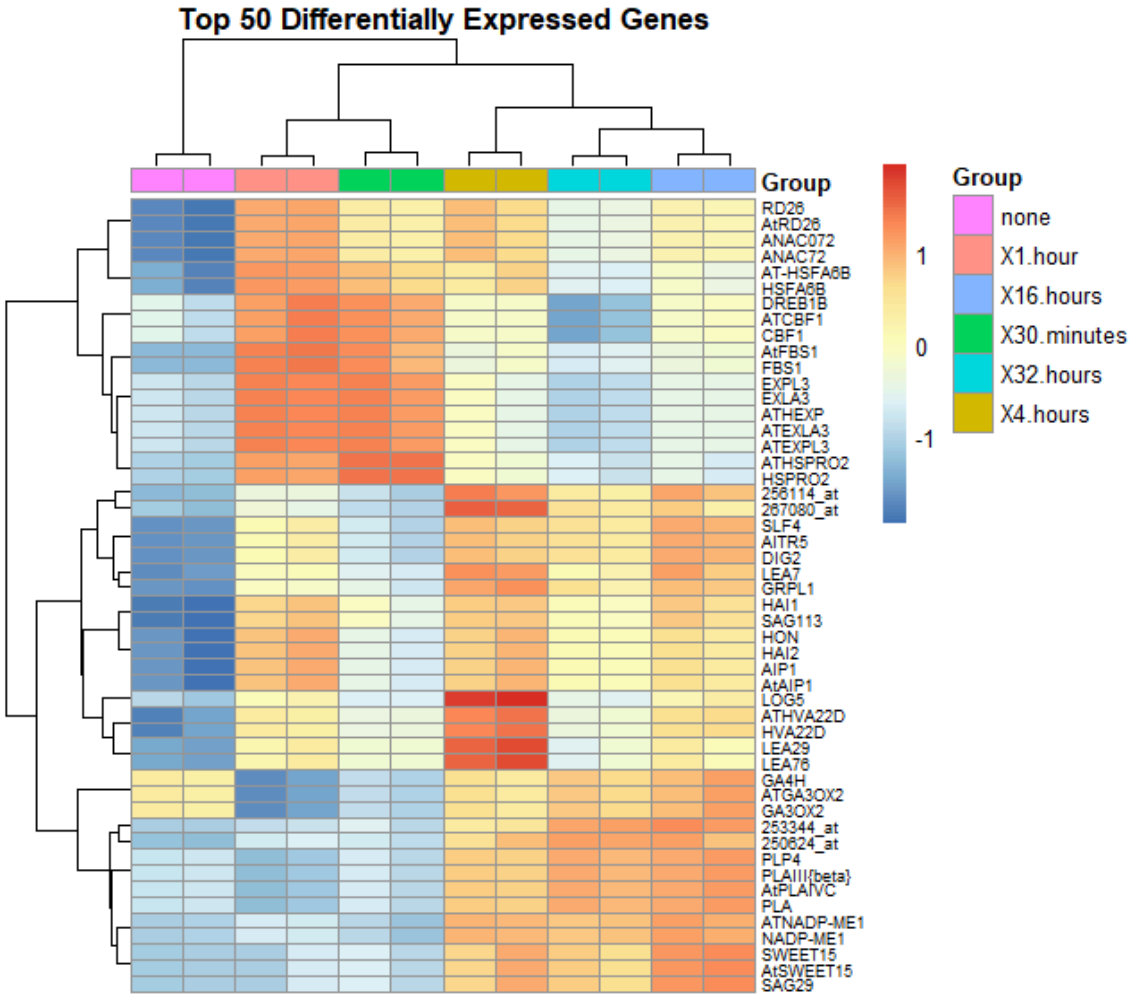

Figure 3. Heatmap of the top 50 DEGs at five time points of salt-stressed *A. thaliana* and control from the GSE7642 dataset. The color represents a range of upregulated (in red) and downregulated (in blue) genes.

Table 2. List of the top five significantly (p-value < 0.05) annotated GO terms. The list was generated by ShinyGo from the submitted top 46 DEGs of GSE7642.

FDR	Associated DEGs	GO ID	GO Term
5.85E-05	LEA7, HVA22D, CBF1, RD26, HAI1	GO:0009414	Response to water deprivation
5.85E-05	FBS1, HVA22D, RD26, PLP4, HAI1	GO:0009737	Response to abscisic acid
3.89E-04	HAI2, FBS1, GA3OX2, HSPRO2, CBF1, SAG29, HAI1	GO:0005515	Protein binding
1.50E-03	HAI2, HAI1	GO:0009788	Negative regulation of abscisic acid-activated signaling pathway
2.13E-03	HAI2, LEA7, NADP-ME1, LOG5, HAI1	GO:0005829	Cytosol

Top five biological processes (FDR < 0.05) that correlated with the submitted DEGs were related to response to abscisic acid (GO:0009737), response to water deprivation (GO:0009414), protein binding (GO:0005515), negative regulation of abscisic acid-activated signaling pathway (GO:0009788), and cytosol (GO:0005829). Abscisic acid (ABA) is a growth-inhibiting phytohormone that is responsible for water-limiting abiotic stress, including salinity. ABA can induce the closing of plant stomata to minimize water loss (Lamers et al., 2025; Liu et al., 2024). According to Liu et al. (2024), ABA cascades in salt stress signals can be classified into ABA-dependent and ABA-independent. There are three protein families that act as key factors: PYR/PYL/RCARs as an ABA receptor, protein phosphatase 2C (PP2Cs) as negative regulators, and SNF1-related protein kinase 2s (SnRK2s) as positive regulators of downstream signals.

Responsive to desiccation 26 (RD26) and F-boxes stress induced 1 (FBS1) are two of the earliest genes that are upregulated after 30 minutes of salt stress exposure. RD26 will improve plants' tolerance to abiotic stress by encoding the NAC-type of transcription factor (TF) that activates ABA-dependent signaling (Fujita et al., 2004). FBS1 also increases plant survival in saline conditions and limits root growth (Geem et al., 2025). Highly ABA-induced 1 (HAI1) and 2 (HAI2) are members of the PP2C family that are upregulated after 1 hour of salt stress exposure. They are found to play a role in negative regulators of osmotic adjustment (Bhaskara et al., 2012). HVA22D also upregulated after 1 hour and peaked in 4 hours of salt-stress exposure. HVA22D promotes salt tolerance by regulating autophagy (Chen et al., 2009). Patatin-like protein 4 (PLP4) started to upregulate after 4 hours of salt stress exposure. PLP4 is a member of the patatin-related phospholipase A (PLA) family. PLA family is essential in lipid metabolism and stress signaling to regulate water status in an ABA-independent manner (Yang et al., 2012).

D. Conclusion

In conclusion, the highest DEGs of 824 were successfully identified with R in the 4-hour time point of salt stress compared to normal conditions from the GSE7642 dataset. The 46 top significant genes were found to be significantly related to abscisic acid and water deprivation responses.

E. References

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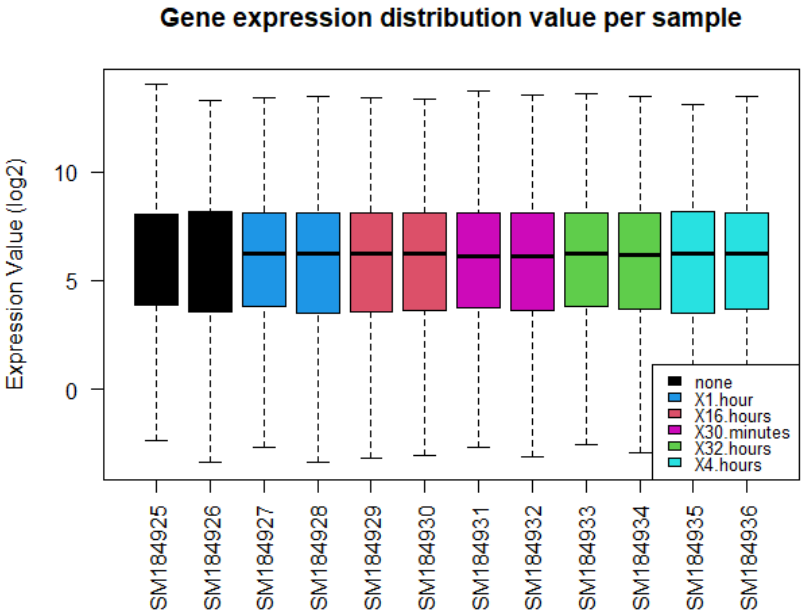
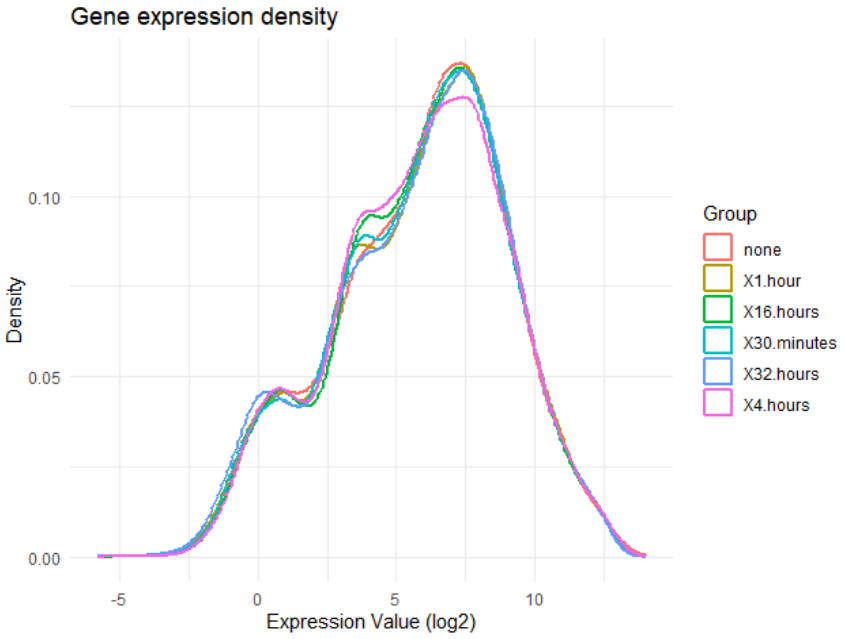
F. Supplementary



Supplementary 1. Boxplot of GSE7642 dataset. The plot shows the gene expression distribution value of each sample from the dataset.



Supplementary 2. Density plot of GSE7642 dataset. The plot shows a left-skewed (long tail to the left side of the graph) and multimodal (more than one distinct peak) data density of the gene expression in each grouping.